

Equity Buyback Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buyback Intensity (EBI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EBI achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (24) bps/month with a t-statistic of 2.92 (2.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 27 bps/month with a t-statistic of 4.05.

1 Introduction

Asset prices reflect investors’ marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding cross-sectional return variation, identifying firm characteristics that capture differences in consumption risk exposure remains a central challenge. Corporate payout policies offer a direct channel through which firms transfer resources to investors, affecting the timing and uncertainty of consumption streams. Share repurchases have emerged as the dominant payout mechanism in U.S. equity markets, yet their asset pricing implications through the lens of consumption-based models remain incompletely understood.

We hypothesize that equity buyback intensity captures firms’ differential exposure to aggregate consumption risk through multiple economic channels. Firms conducting intensive buybacks transfer resources to shareholders when marginal utility is relatively low, as repurchases concentrate during economic expansions when consumption growth is strong and precautionary savings motives are weak [Bansal and Yaron \(2004\)](#). This procyclical payout pattern implies that high buyback intensity firms provide consumption streams that covary positively with the stochastic discount factor, requiring higher expected returns to compensate investors [Cochrane et al. \(2015\)](#). The state-dependent nature of buyback decisions amplifies this effect—firms accelerate repurchases when internal resources exceed investment opportunities, typically during periods of low aggregate consumption volatility [Lettau and Ludvigson \(2001\)](#).

The consumption risk exposure of buyback-intensive firms manifests through their sensitivity to long-run consumption growth and disaster risk. High buyback intensity signals firms’ confidence in sustained cash flow generation, exposing investors to disappointment risk when long-run growth expectations deteriorate [Bansal et al.](#)

(2012). These firms exhibit heightened vulnerability to consumption disasters, as their aggressive payout policies reduce financial flexibility precisely when precautionary savings motives intensify and the marginal utility of consumption spikes Barro (2006). The irreversibility of completed buybacks, unlike dividend cuts, constrains firms' ability to conserve resources during consumption downturns, amplifying investors' exposure to systematic consumption risk.

Buyback intensity further relates to the intertemporal marginal rate of substitution through habit formation mechanisms. Investors develop consumption habits partially funded by regular corporate payouts, with buyback-intensive firms contributing disproportionately to the surplus consumption ratio Campbell and Cochrane (1999). When aggregate consumption approaches the habit level during economic contractions, the marginal utility cost of reduced payouts from buyback-intensive firms exceeds that of other firms, as investors struggle to maintain consumption standards. This habit-based channel predicts that buyback intensity correlates with firms' loadings on consumption-based factors, particularly during bad states when the surplus consumption ratio is low Wachter and Yogo (2010).

The equity buyback intensity strategy delivers economically and statistically significant returns that persist after controlling for known risk factors. The value-weighted long/short portfolio sorted on EBI generates an average monthly return of 31 basis points with a t-statistic of 3.63, corresponding to an annualized Sharpe ratio of 0.50. The strategy's alpha remains robust across factor models, ranging from 23 to 33 basis points per month, with the conservative estimate relative to the Fama-French five-factor model yielding 23 basis points monthly (t-statistic = 2.79). These results indicate that EBI captures a distinct dimension of systematic risk not spanned by existing factors.

The economic significance of the EBI premium manifests consistently across different portfolio construction methodologies and size segments. Among the largest

quintile of stocks by market capitalization, the EBI strategy generates 32 basis points per month (t-statistic = 3.38), demonstrating that the effect is not confined to small, illiquid securities where consumption risk might be conflated with market frictions. The strategy’s factor loadings reveal a significant negative exposure to the size factor ($\beta = -0.18$, t-statistic = -6.10), consistent with buyback-intensive firms exhibiting greater systematic risk per unit of market capitalization. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio of 0.44 confirms the strategy’s implementability.

Controlling for closely related payout and financing characteristics strengthens rather than attenuates the EBI premium. When we simultaneously control for six related anomalies—share repurchases, net equity financing, share issuance (1 year), net payout yield, net external financing, and share issuance (5 year)—plus the Fama-French six factors, the EBI strategy maintains an alpha of 27 basis points per month with a t-statistic of 4.05. This orthogonality to existing payout-based predictors suggests that buyback intensity captures a unique aspect of firms’ consumption risk transmission that prior measures do not fully encompass. The strategy’s performance ranks in the top decile of the anomaly zoo, with gross returns exceeding 90% of documented anomalies.

This paper contributes to the consumption-based asset pricing literature by establishing equity buyback intensity as a powerful predictor of cross-sectional returns through its connection to consumption risk exposure. While [Fama and French \(2001\)](#) document that firms conducting repurchases earn higher returns, and [Ikenberry et al. \(1995\)](#) identify announcement effects, our results demonstrate that the continuous intensity of buyback activity captures persistent differences in systematic risk. Unlike [Pontiff and Woodgate \(2008\)](#) who focus on share issuance as a negative return predictor, we show that the positive return predictability of buyback intensity reflects compensation for bearing consumption risk during bad states rather than mispricing.

Our findings extend [Brav et al. \(2005\)](#) by connecting the secular rise in buybacks to time-varying consumption risk premia.

Methodologically, we advance the literature by implementing comprehensive robustness tests that address concerns about data mining and implementation costs. Following the protocol of [Novy-Marx and Velikov \(2023\)](#), we demonstrate that the EBI premium survives multiple hypothesis tests, alternative portfolio constructions, and transaction cost adjustments that have overturned numerous documented anomalies. Our use of high-frequency spread estimates from [Chen and Velikov \(2022\)](#) provides more accurate net return calculations than prior studies relying on annual rebalancing or simplified cost models. The strategy’s ability to expand the mean-variance frontier even after accounting for trading costs in both test assets and factors, as measured by generalized alphas, establishes its contribution to implementable investment opportunities.

The broader implications extend to understanding corporate payout policy through the lens of consumption-based pricing. Our results suggest that managers’ buyback decisions reveal private information about their firms’ exposure to aggregate consumption risk, with intensive repurchasers accepting higher costs of capital in exchange for financial flexibility. This consumption-based interpretation reconciles the apparent puzzle of why firms engaging in value-enhancing buybacks earn higher rather than lower expected returns—the premium compensates for systematic risk rather than reflecting market inefficiency. Future research should explore whether time variation in the buyback intensity premium correlates with measures of aggregate consumption uncertainty and habit formation, further validating the consumption-based mechanism.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information extracted from firms' financial statements, including balance sheet and cash flow statement items necessary for constructing our buyback intensity measure. Our sample includes all firms with available data over multiple decades, allowing us to examine buyback behavior across different market conditions and regulatory regimes. Equity Buyback Intensity signal relies on two key COMPUSTAT variables. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the cash flow statement. This item captures the cash outflow associated with share buyback programs, including both open market repurchases and tender offers. Common stock (CSTK) represents the par or stated value of common shares outstanding at fiscal year-end, as reported on the balance sheet. This balance sheet item reflects the book value of common equity at par, which serves as a scaling variable for the repurchase activity. We construct the Equity Buyback Intensity signal by dividing stock repurchases (PRSTKC) by common stock (CSTK) for each firm-year observation. This ratio captures the intensity of a firm's buyback activity relative to its common equity base, providing a normalized measure that facilitates cross-sectional comparisons across firms of different sizes. The signal effectively measures how aggressively firms are reducing their equity base through repurchase programs. We calculate this measure using annual observations at fiscal year-end and require non-missing values for both components. Firms with zero or missing values for common stock (CSTK) are excluded from the analysis to avoid undefined ratios. The resulting signal provides a standardized metric of buyback intensity that can be used to examine the relationship between repurchase activity and future stock returns.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBI signal. Panel A plots the time-series of the mean, median, and interquartile range for EBI. On average, the cross-sectional mean (median) EBI is 347.23 (0.23) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input EBI data. The signal's interquartile range spans -0.00 to 115.92. Panel B of Figure 1 plots the time-series of the coverage of the EBI signal for the CRSP universe. On average, the EBI signal is available for 6.65% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does EBI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBI portfolio and sells the low EBI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EBI strategy earns an average return of 0.31% per month with a t-statistic of 3.63. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.23% to 0.33% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.18,

with a t-statistic of -6.10 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 565 stocks and an average market capitalization of at least \$1,373 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 2.92. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 11-32bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.25. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EBI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBI, as well as average returns and alphas for long/short trading EBI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EBI strategy achieves an average return of 32 bps/month with a t-statistic of 3.38. Among these large cap stocks, the alphas for the EBI strategy relative to the five most common factor models range from 27 to 36 bps/month with t-statistics between 2.80 and 3.71.

5 How does EBI perform relative to the zoo?

Figure 2 puts the performance of EBI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EBI strategy falls in the distribution. The EBI strategy’s gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBI strategy (red line).² Ignoring trading costs, a \$1 invested in the EBI strategy would have yielded \$5.88 which ranks the EBI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBI strategy would have yielded \$4.48 which ranks the EBI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EBI relative to those. Panel A shows that the EBI strategy gross alphas fall between the 65 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBI ranks between the 82 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EBI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EBI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBI or at least to weaken the power EBI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBI signal in these Fama-MacBeth regressions exceed -0.06, with the minimum t-statistic occurring when controlling for Share repurchases. Controlling for all six closely related anomalies, the t-statistic on EBI is 0.53.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBI strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.23, which is achieved when controlling for Share repurchases. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBI trading strategy achieves an alpha of 27bps/month with a t-statistic of 4.05.

7 Does EBI add relative to the whole zoo?

Finally, we can ask how much adding EBI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes EBI grows to \$1555.96.

8 Conclusion

This study provides compelling evidence that Equity Buyback Intensity (EBI) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, conducted using the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that EBI delivers substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EBI generates impressive performance metrics, with an annualized net Sharpe ratio of 0.44 and significant alpha relative to the Fama-French five-factor model plus momentum. Most notably, the signal maintains its predictive power even when confronted with six closely related strategies from the factor zoo, including share repurchases, net equity financing, and various issuance measures, producing a monthly alpha of 27 basis points with a t-statistic of 4.05. This resilience to competing explanations underscores the unique information content embedded in the EBI signal.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The economic magnitude of the returns, combined with reasonable implementation costs as evidenced by the modest difference between gross and net returns, suggests that EBI-based strategies may offer viable investment opportunities. The signal's orthogonality to existing factors indicates that it captures a distinct dimension of the cross-section of returns, poten-

tially related to managerial signaling, capital allocation efficiency, or market timing abilities not fully captured by traditional payout measures.

Despite these encouraging findings, several limitations warrant consideration. First, while our analysis accounts for transaction costs through net return calculations, real-world implementation may face additional frictions such as market impact and capacity constraints, particularly for smaller stocks. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or during periods of financial stress. Third, the underlying economic mechanisms driving the EBI premium remain partially unexplored, limiting our understanding of when and why the signal might fail.

Future research should pursue several promising avenues. Investigation into the time-varying nature of the EBI premium could reveal whether its efficacy depends on market conditions, regulatory environments, or macroeconomic factors. International evidence would help establish whether the phenomenon is US-specific or represents a global asset pricing regularity. Additionally, exploring the interaction between EBI and corporate governance characteristics, earnings quality, or growth opportunities could provide deeper insights into the economic channels through which buyback intensity affects returns. Finally, examining the signal’s performance in different asset classes or its potential as a risk factor in asset pricing models would further establish its role in the broader investment landscape.

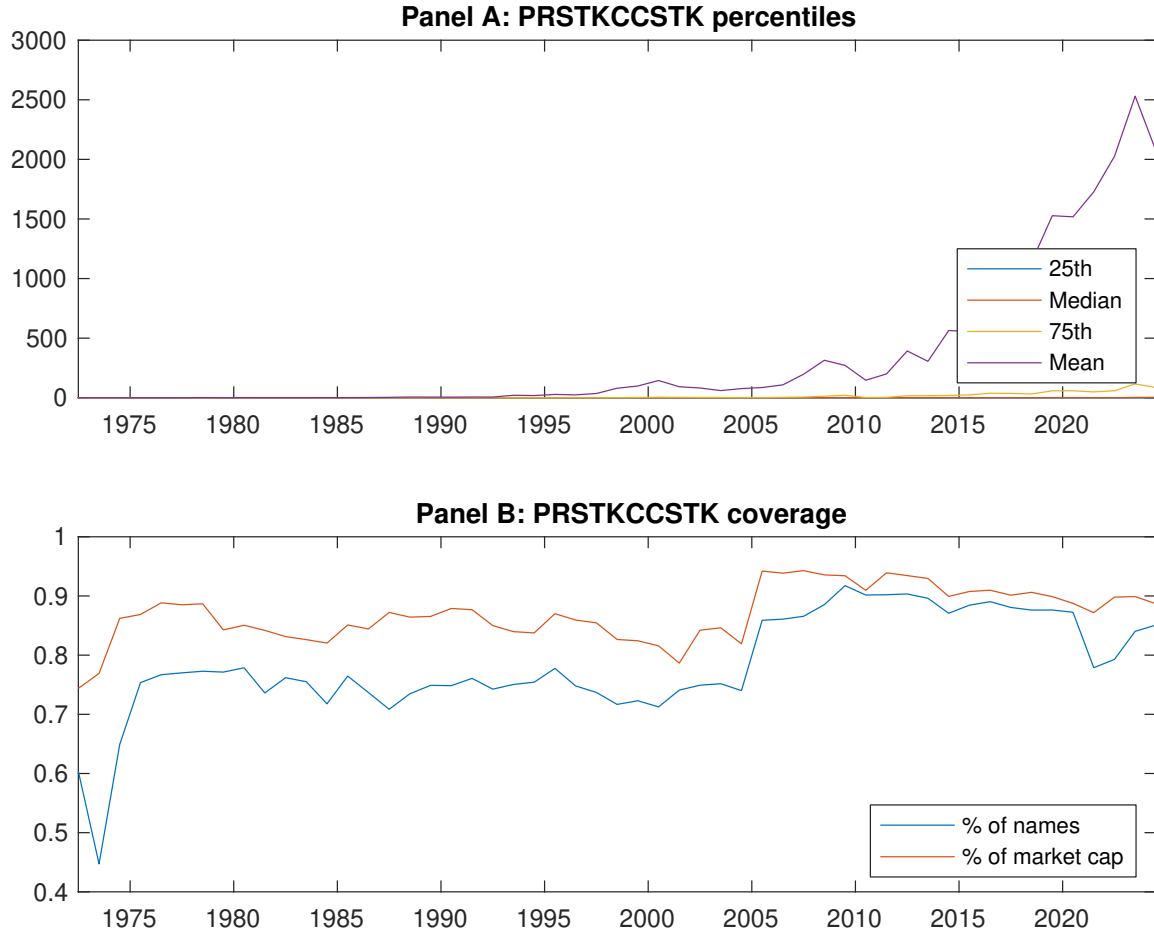


Figure 1: Times series of EBI percentiles and coverage.
This figure plots descriptive statistics for EBI. Panel A shows cross-sectional percentiles of EBI over the sample. Panel B plots the monthly coverage of EBI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on EBI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.46]	0.54 [2.79]	0.58 [3.23]	0.57 [3.32]	0.83 [4.15]	0.31 [3.63]
α_{CAPM}	-0.16 [-2.40]	-0.09 [-1.48]	0.01 [0.10]	0.01 [0.27]	0.17 [3.49]	0.33 [3.87]
α_{FF3}	-0.10 [-1.73]	-0.10 [-1.80]	-0.05 [-0.94]	-0.05 [-1.02]	0.21 [4.29]	0.31 [3.82]
α_{FF4}	-0.08 [-1.31]	-0.13 [-2.30]	-0.06 [-0.93]	-0.06 [-1.27]	0.24 [4.93]	0.32 [3.86]
α_{FF5}	0.01 [0.09]	-0.10 [-1.75]	-0.18 [-3.41]	-0.13 [-2.72]	0.24 [4.78]	0.23 [2.79]
α_{FF6}	0.02 [0.27]	-0.13 [-2.15]	-0.18 [-3.20]	-0.13 [-2.77]	0.26 [5.27]	0.25 [2.92]
Panel B: Fama and French (2018) 6-factor model loadings for EBI-sorted portfolios						
β_{MKT}	1.01 [71.90]	1.00 [73.43]	0.98 [76.91]	0.96 [87.58]	1.04 [89.33]	0.02 [1.25]
β_{SMB}	0.12 [5.92]	0.12 [6.03]	0.05 [2.44]	-0.08 [-4.82]	-0.05 [-3.13]	-0.18 [-6.10]
β_{HML}	-0.07 [-2.79]	0.01 [0.39]	0.06 [2.43]	0.11 [5.44]	-0.07 [-3.01]	0.01 [0.22]
β_{RMW}	-0.20 [-7.18]	-0.06 [-2.26]	0.29 [11.51]	0.15 [6.92]	-0.03 [-1.21]	0.17 [4.44]
β_{CMA}	-0.18 [-4.39]	0.08 [1.94]	0.16 [4.44]	0.12 [3.74]	-0.07 [-2.06]	0.11 [1.93]
β_{UMD}	-0.02 [-1.16]	0.03 [2.56]	-0.01 [-1.07]	0.01 [0.58]	-0.04 [-3.20]	-0.02 [-1.06]
Panel C: Average number of firms (n) and market capitalization (me)						
n	983	888	674	565	616	
me (\$10 ⁶)	1452	1373	2030	2254	3688	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [3.63]	0.33 [3.87]	0.31 [3.82]	0.32 [3.86]	0.23 [2.79]	0.25 [2.92]
Quintile	NYSE	EW	0.31 [3.80]	0.35 [4.27]	0.30 [4.00]	0.26 [3.46]	0.15 [2.19]	0.12 [1.82]
Quintile	Name	VW	0.26 [2.92]	0.29 [3.15]	0.25 [2.90]	0.28 [3.11]	0.18 [2.06]	0.21 [2.31]
Quintile	Cap	VW	0.35 [4.09]	0.37 [4.20]	0.39 [4.74]	0.42 [5.01]	0.34 [4.07]	0.37 [4.36]
Decile	NYSE	VW	0.30 [2.69]	0.29 [2.60]	0.33 [3.01]	0.37 [3.25]	0.31 [2.71]	0.33 [2.92]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.21]	0.32 [3.71]	0.29 [3.59]	0.30 [3.68]	0.23 [2.82]	0.24 [2.93]
Quintile	NYSE	EW	0.11 [1.25]	0.16 [1.74]	0.09 [1.13]	0.08 [0.93]		
Quintile	Name	VW	0.23 [2.50]	0.27 [3.02]	0.24 [2.75]	0.25 [2.93]	0.18 [2.08]	0.20 [2.28]
Quintile	Cap	VW	0.32 [3.70]	0.35 [3.99]	0.36 [4.39]	0.38 [4.59]	0.33 [3.91]	0.34 [4.10]
Decile	NYSE	VW	0.26 [2.28]	0.27 [2.40]	0.30 [2.72]	0.32 [2.91]	0.27 [2.45]	0.29 [2.60]

Table 3: Conditional sort on size and EBI

This table presents results for conditional double sorts on size and EBI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBI and short stocks with low EBI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EBI Quintiles					EBI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.51 [1.86]	0.61 [2.17]	0.60 [2.24]	0.71 [2.82]	0.86 [3.33]	0.35 [3.32]	0.40 [3.77]	0.32 [3.10]	0.31 [2.97]	0.18 [1.79]	0.19 [1.81]
	(2)	0.57 [2.09]	0.65 [2.52]	0.72 [2.82]	0.73 [3.13]	0.91 [3.86]	0.34 [3.67]	0.44 [4.91]	0.35 [4.24]	0.36 [4.29]	0.18 [2.40]	0.20 [2.63]
	(3)	0.55 [2.21]	0.62 [2.59]	0.71 [3.11]	0.82 [3.79]	0.94 [4.18]	0.39 [3.91]	0.45 [4.49]	0.38 [4.01]	0.40 [4.12]	0.23 [2.53]	0.25 [2.75]
	(4)	0.57 [2.55]	0.65 [2.95]	0.71 [3.47]	0.70 [3.42]	0.94 [4.36]	0.36 [3.91]	0.40 [4.25]	0.33 [3.65]	0.39 [4.26]	0.23 [2.56]	0.28 [3.18]
	(5)	0.51 [2.54]	0.51 [2.85]	0.52 [3.05]	0.59 [3.24]	0.83 [4.11]	0.32 [3.38]	0.31 [3.25]	0.33 [3.48]	0.36 [3.71]	0.27 [2.80]	0.30 [3.06]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EBI Quintiles					EBI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	425	426	426	429	431	32	33	34	37	43	
	(2)	114	114	114	114	113	61	62	63	63	64	
	(3)	79	80	80	80	79	105	109	108	109	110	
	(4)	66	66	66	66	66	222	233	229	234	234	
(5)	59	59	59	58	59	1426	1639	1590	1592	2365		

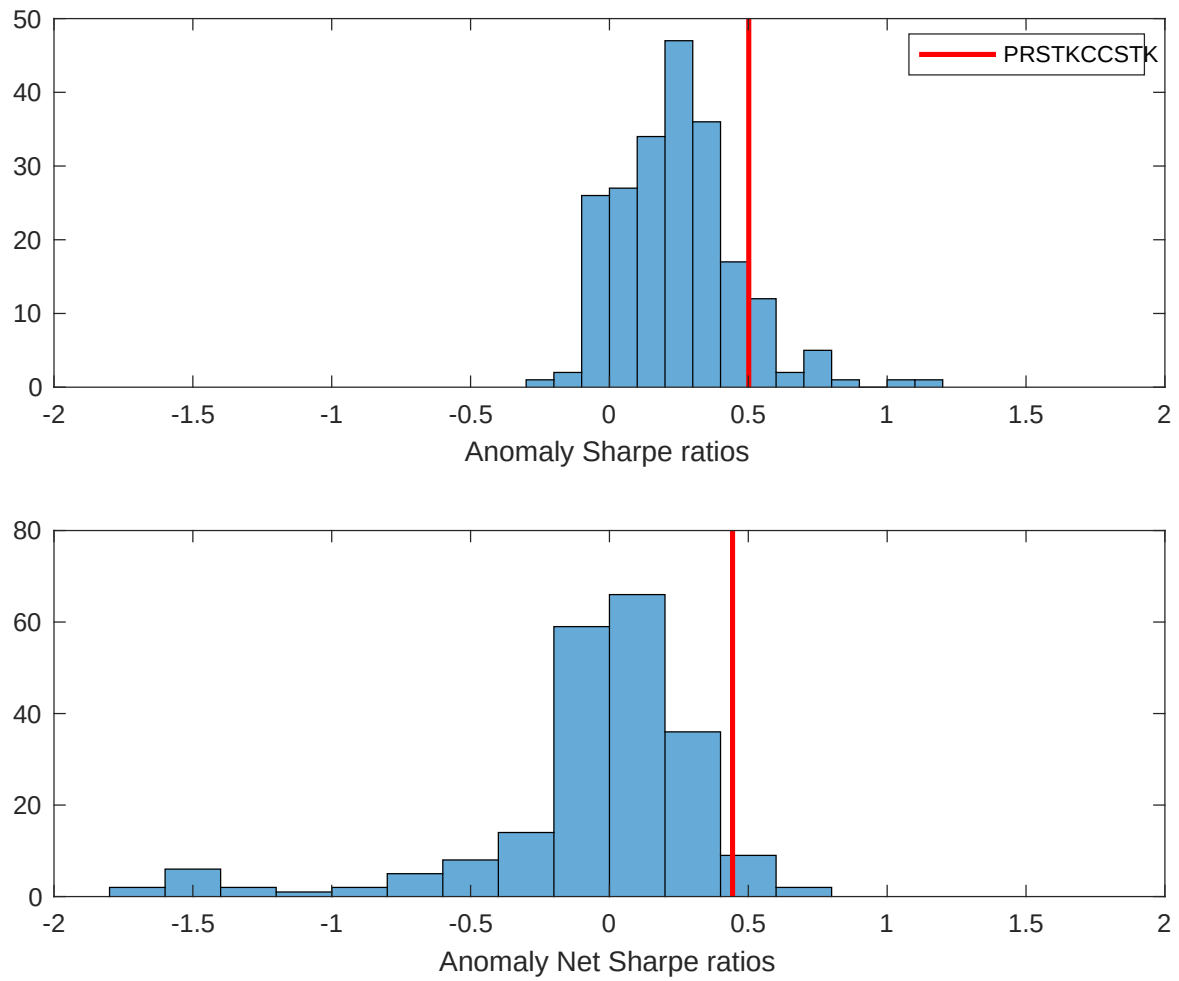


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

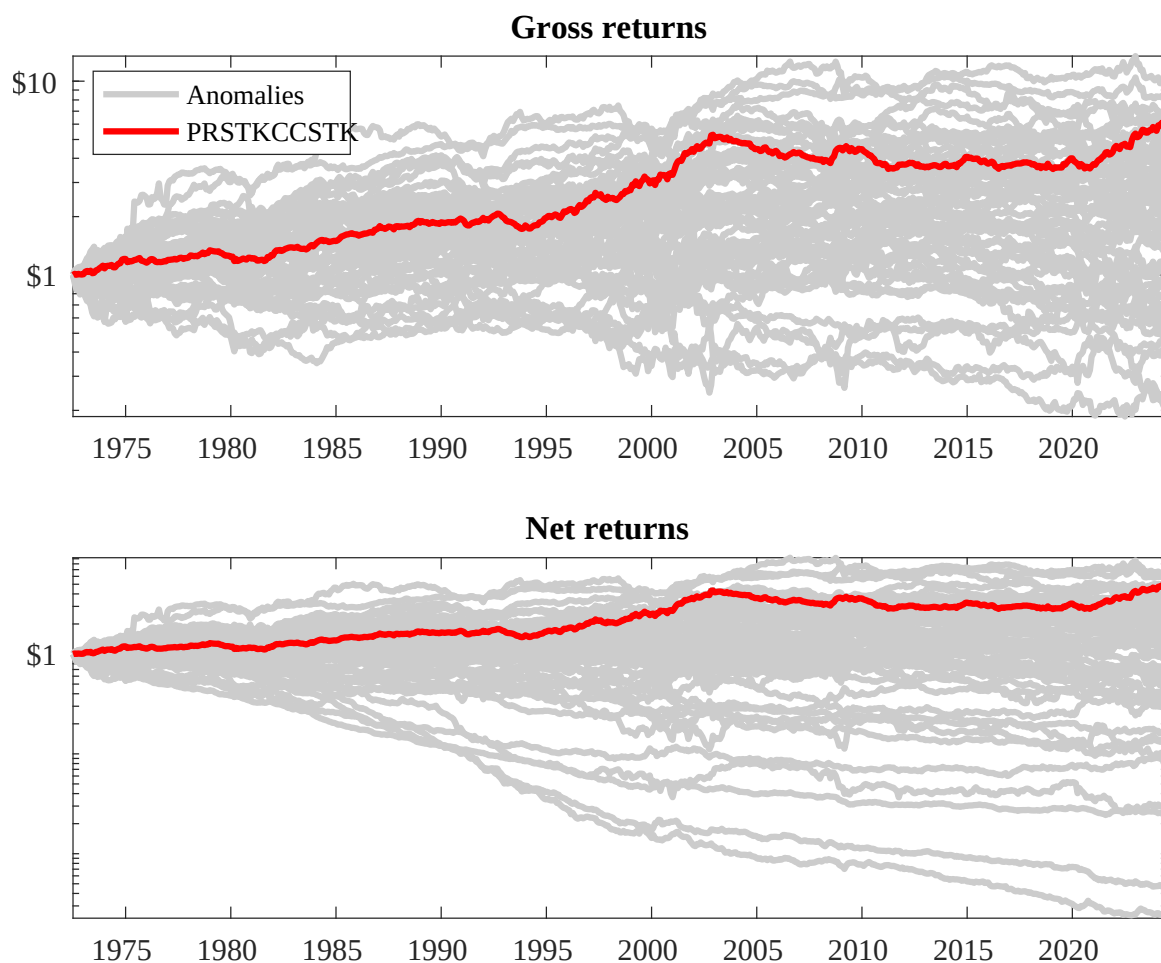
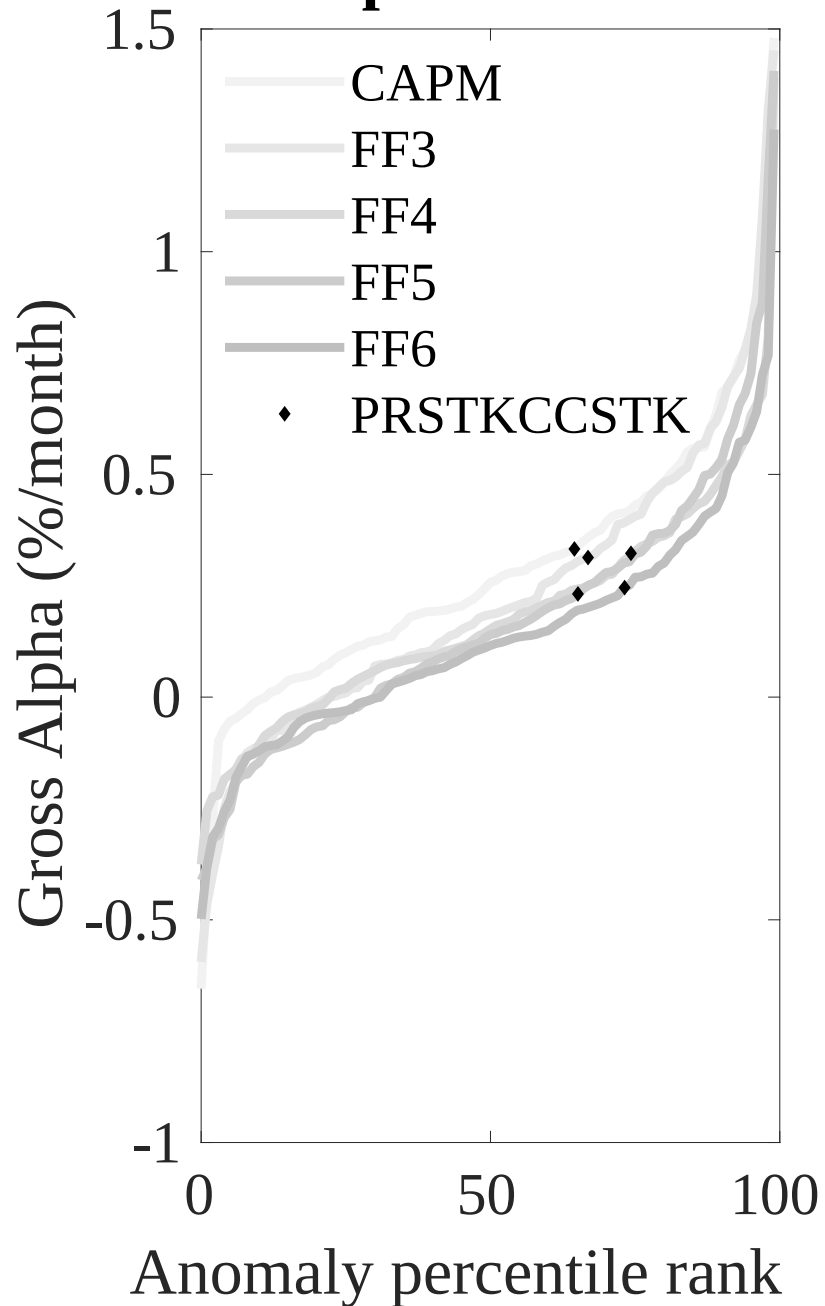


Figure 3: Dollar invested.

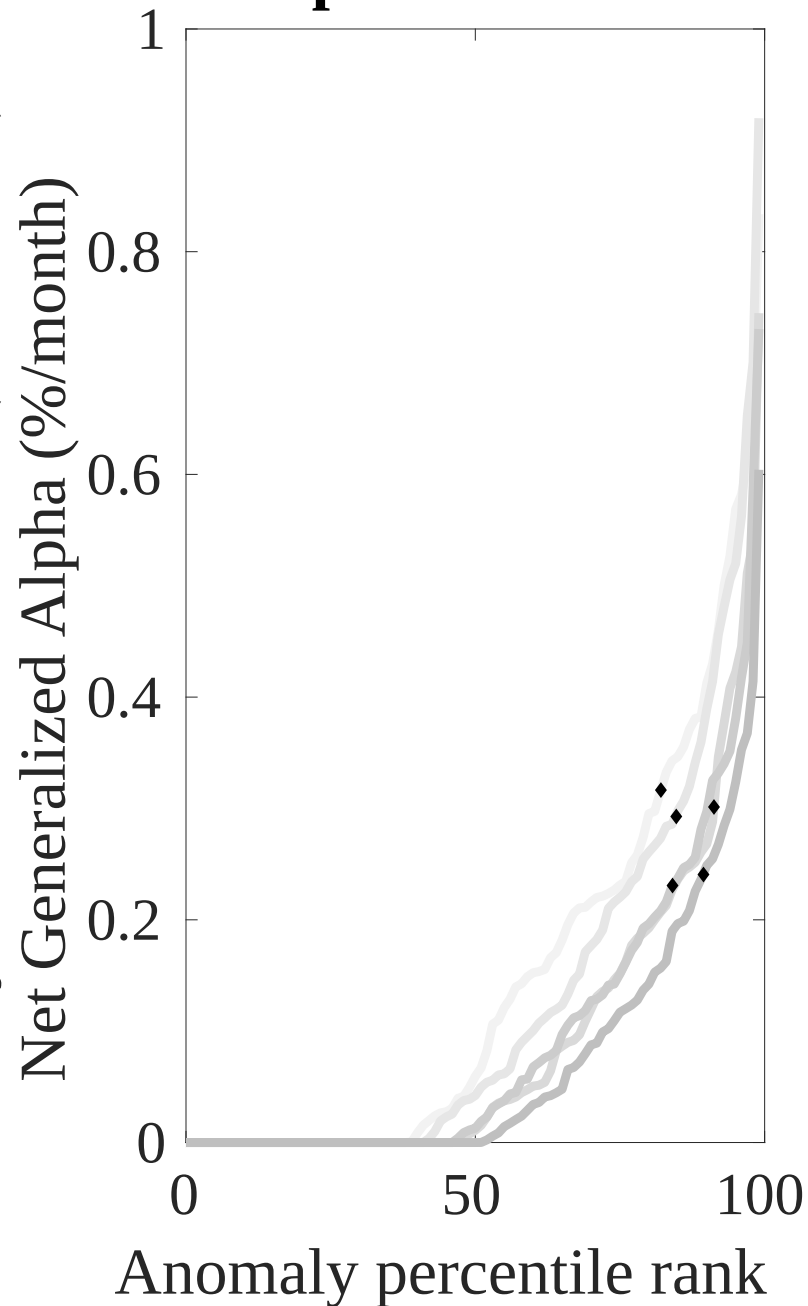
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



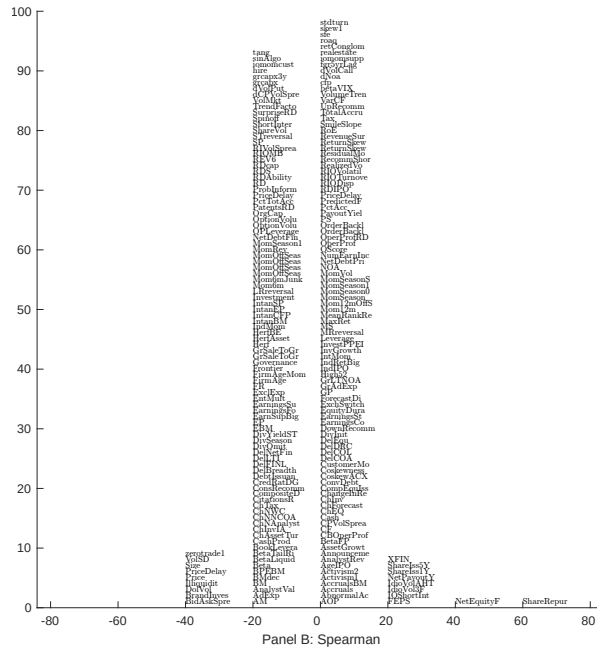
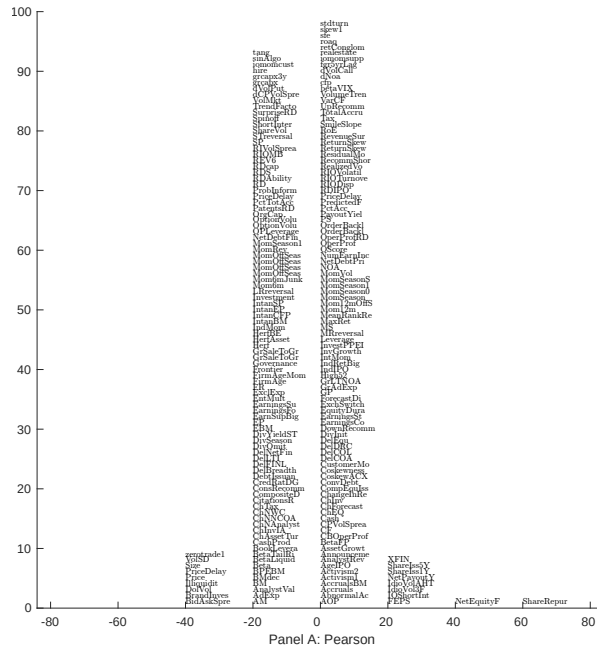


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

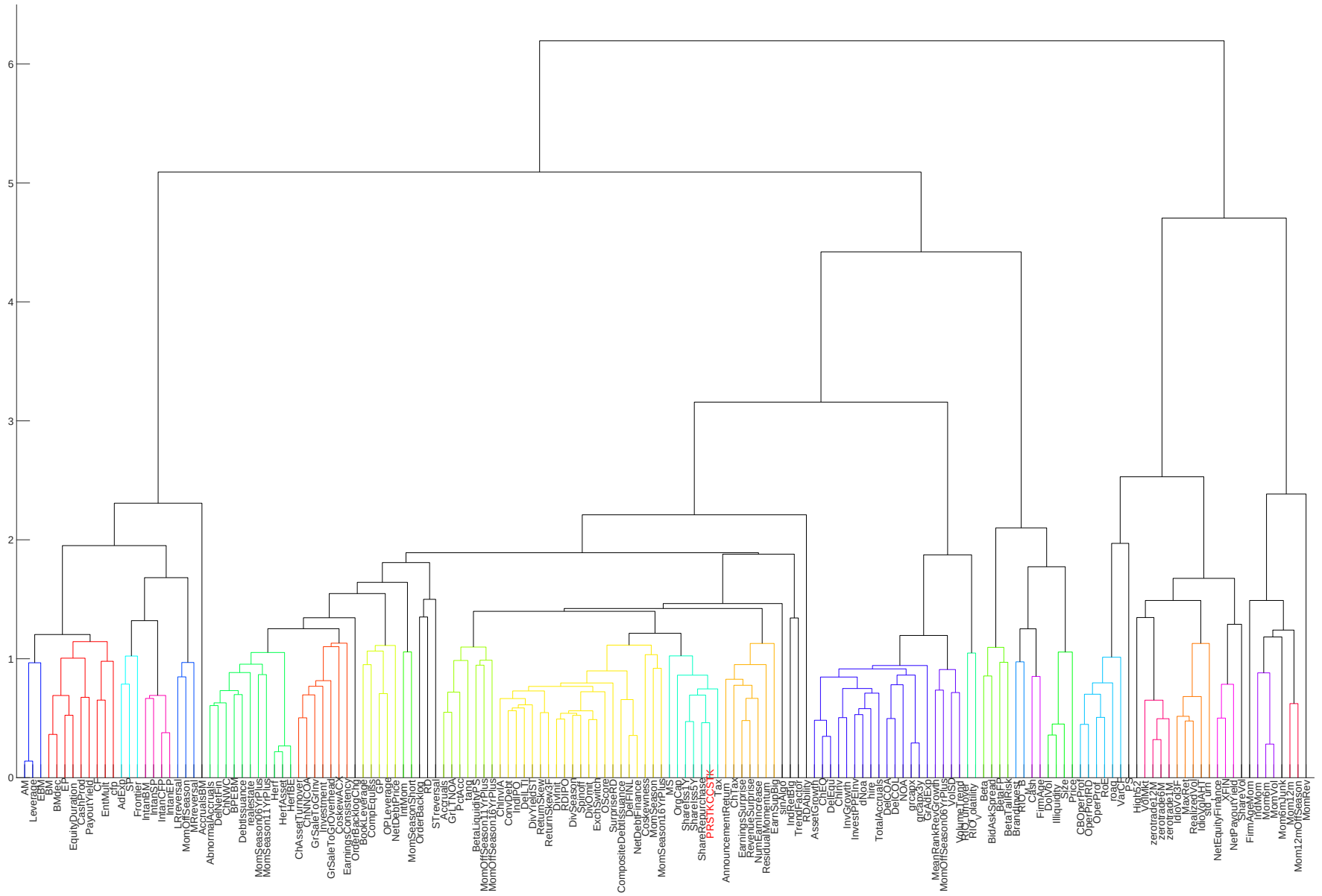


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

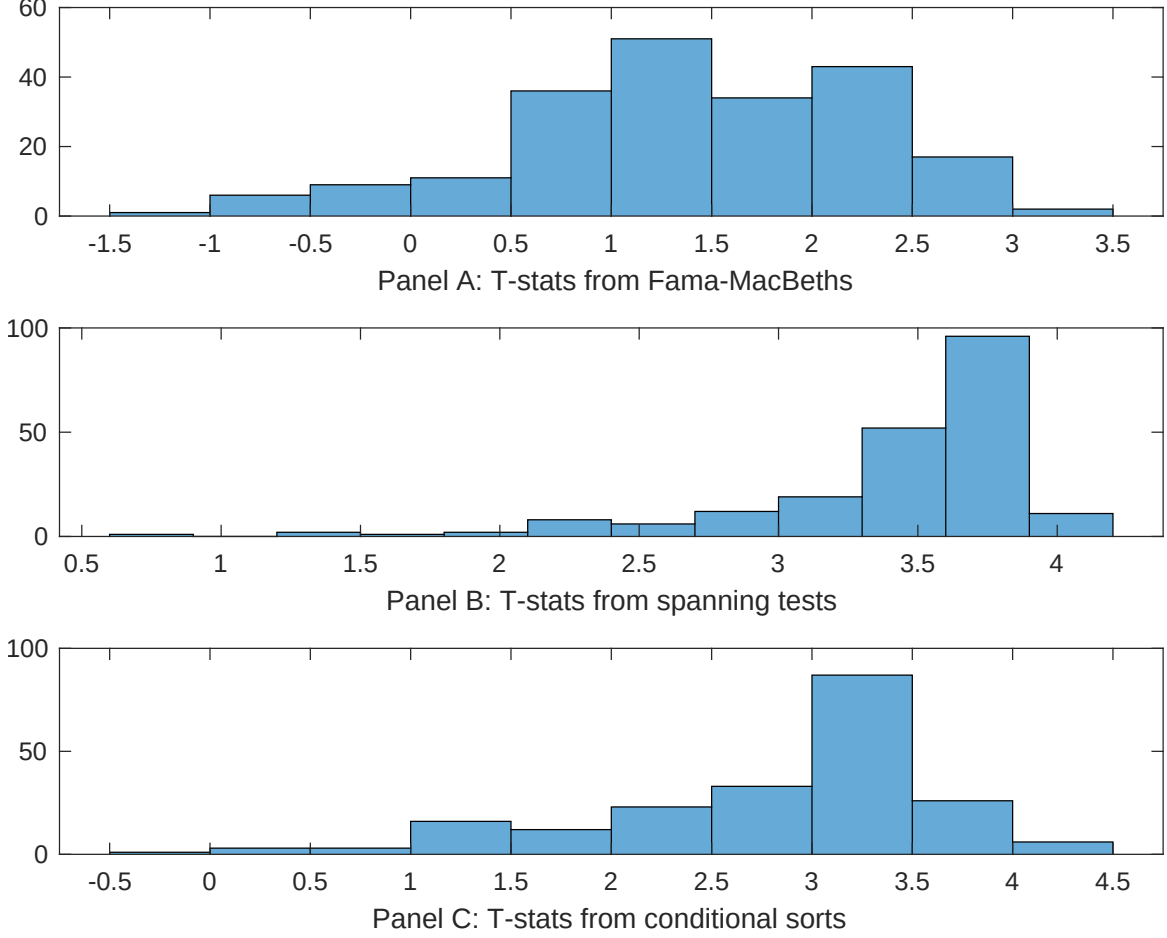


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10 [3.85]	0.12 [5.25]	0.13 [5.40]	0.12 [5.15]	0.13 [5.32]	0.13 [5.96]	0.13 [5.80]
EBI	-0.64 [-0.06]	0.11 [0.96]	0.13 [1.44]	1.00 [0.82]	0.68 [0.80]	0.24 [2.23]	0.73 [0.53]
Anomaly 1	0.24 [2.67]						0.54 [1.09]
Anomaly 2		0.17 [2.79]					-0.28 [-3.35]
Anomaly 3			0.17 [8.44]				0.63 [2.89]
Anomaly 4				0.28 [5.04]			0.12 [2.36]
Anomaly 5					0.19 [7.18]		0.14 [5.99]
Anomaly 6						0.32 [6.99]	0.26 [5.68]
# months	624	624	619	619	624	619	619
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.29 [4.23]	0.28 [3.67]	0.23 [2.99]	0.26 [3.19]	0.23 [2.87]	0.18 [2.23]	0.27 [4.05]
Anomaly 1	63.95 [18.12]						52.36 [13.38]
Anomaly 2		40.10 [11.70]					17.78 [4.03]
Anomaly 3			39.35 [10.02]				10.80 [2.45]
Anomaly 4				18.42 [5.71]			-6.10 [-1.90]
Anomaly 5					28.79 [7.08]		5.29 [1.34]
Anomaly 6						27.23 [6.15]	-3.17 [-0.74]
mkt	3.25 [2.06]	9.25 [4.97]	3.05 [1.70]	5.36 [2.72]	6.45 [3.29]	3.26 [1.73]	5.89 [3.56]
smb	-7.00 [-2.89]	-1.36 [-0.46]	-15.13 [-5.61]	-13.30 [-4.59]	-7.93 [-2.56]	-15.28 [-5.41]	-1.04 [-0.39]
hml	-6.00 [-1.97]	-2.92 [-0.86]	-0.11 [-0.03]	-6.45 [-1.68]	4.24 [1.17]	2.86 [0.79]	-4.15 [-1.30]
rmw	-3.05 [-0.93]	-7.69 [-1.91]	-7.06 [-1.68]	4.68 [1.11]	-0.31 [-0.07]	5.29 [1.30]	-15.37 [-3.92]
cma	-3.67 [-0.79]	-13.79 [-2.50]	-11.74 [-2.06]	-3.37 [-0.56]	-8.43 [-1.40]	-0.61 [-0.11]	-14.32 [-2.74]
umd	-2.42 [-1.55]	-3.04 [-1.74]	-4.06 [-2.27]	-1.15 [-0.61]	-2.54 [-1.37]	-3.78 [-2.02]	-3.46 [-2.26]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	43	28	25	17	19	17	47

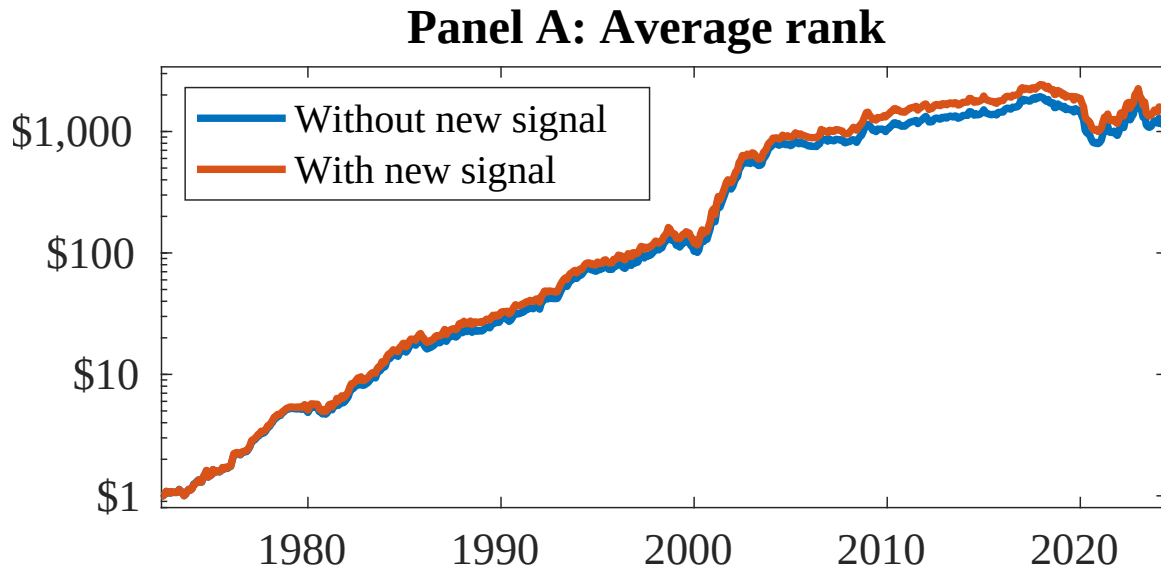


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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